

Vibrating Rod Level Switch

For Solid Applications, Model: VRLS-500S



PRODUCT DESCRIPTION:

The VRLS-500S- Vibrating Rod Level Switch is an economical choice in level detection of powders and bulk solids. The Vibrating Rod Level Switch built with a piezoelectric crystal that vibrates the rod at a natural frequency. Once the rod comes in contact with the material, the vibration will be damped and the switch will change its state. Once the rod is uncovered by the material, the switch will revert to its normal state. Best for silo or hopper applications, its probe design allows for self-cleaning, ensuring no build-up or bridging of material and accurate detection. Sensitivity is adjustable for detection ranging from large granular material to small powders with low bulk densities. The failsafe mode can be set for failure on high level or failure on low level using a selector switch in instrument.

Applications:

Flakes, Grains, Coal Powder, cereals, sugar, animal feed, milk powder, flour, insulation, cement, plastic granules, saw dust, wood chips carbon black, light fibers, detergent powders, dyes, chalk, silica,

SALIENT FEATURES:

- Compact design with integral electronics
- No calibration required – Plug & Play operation
- No moving parts – no wear and tear
- Universal power supply for AC / DC operation
- Relay or Open Collector PNP output options available
- Rigid and sturdy construction for reliable industrial operation
- Available with 250 mm standard rigid probe
- Available with 1½" and above screwed mounting connections
- IP68 ingress protection
- Field-selectable Fail-Safe mode
- LED indication for Normal, Alarm and Status conditions
- Selectable density setting for different bulk solid materials
- Selectable switching delay for stable operation
- Suitable for powders, granules and bulk solids
- Easy installation with minimum maintenance



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OPERATING PRINCIPLE:

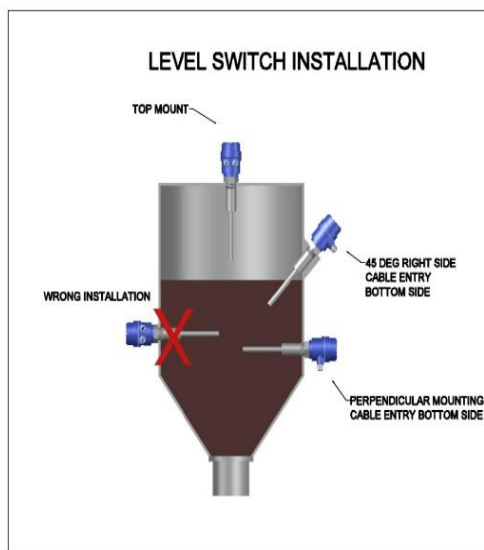
The VFLS-500S Vibrating Rod Level Switch operates on the principle of vibration sensing. The sensing rod is continuously excited at its natural frequency by an electronic oscillator. When the material level is below the sensing rod, the rod continues to vibrate at its normal frequency. When the material reaches and covers the rod, the vibration is damped due to contact with the material. The electronic circuit detects this change in vibration and generates a corresponding level switching signal through the selected relay or PNP output. When the material level falls below the sensing rod, the rod resumes its normal vibration and the output returns to its previous state. The density setting and switching delay can be selected for reliable level detection of different bulk solid materials.

PROBE CONSTRUCTION SUITABLE FOR MATERIAL BUILD-UP:

The VFLS-500S is equipped with a robust vibrating rod probe specifically designed for reliable point level detection in dry and free-flowing bulk solid materials.

The rigid probe construction ensures consistent vibration and stable switching performance when the material comes into contact with the sensing rod. The design is particularly suitable for non-sticky powders, granules and free-flowing solids, where smooth material movement around the probe is required for reliable level detection.

The compact and sturdy probe provides excellent mechanical strength, long service life and dependable operation in industrial silos, hoppers, bins and storage vessels. Suitable for: Powders • Granules • Pellets • Dry Bulk Solids • Free-Flowing Material

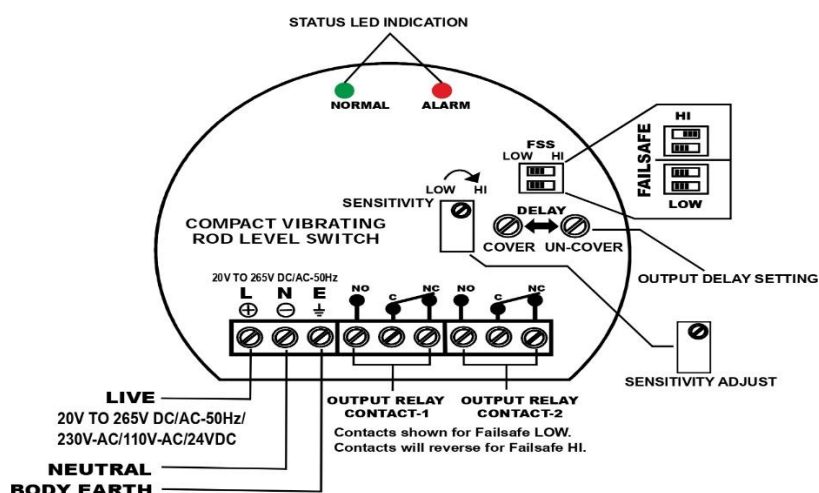


COMPONENTS OF COMPACT LEVEL SWITCH ELECTRICAL CONNECTION ARE EXPLAINED IN FIGURE



- ----Aluminum Housing
- -----Cable Entry
- -----Mounting Thread
- -----Extension pipe
- ----- Tuning Rod

UNIVERSAL /230 VAC/110VAC/24 VDC ELECTRICAL CONNECTION- UNIVERSAL





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DATA SHEET

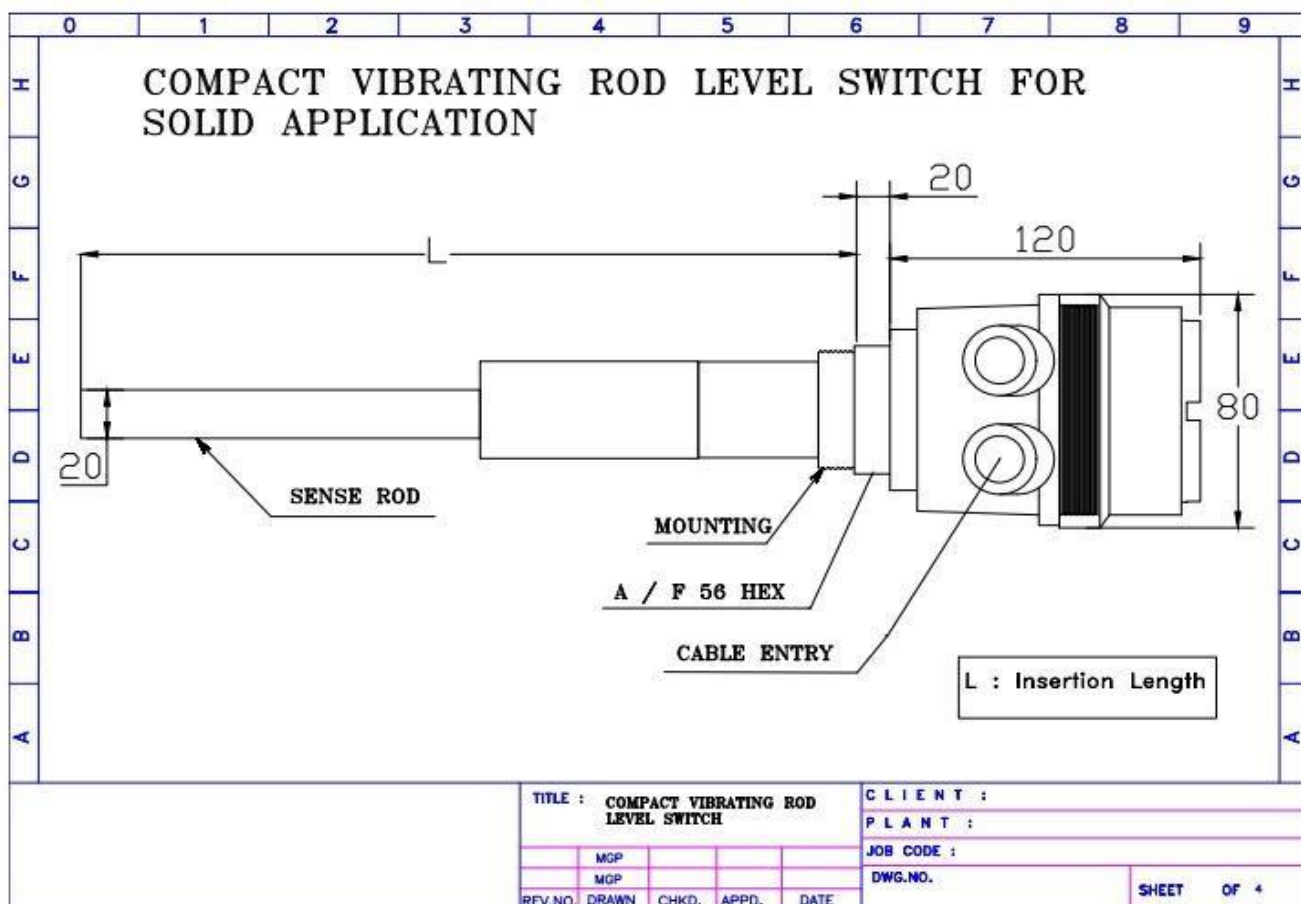
VRLS-500S Vibrating Rod Level Switch Selection / Ordering Information

Selection	Code	Description
Model No.	VRLS-500	Vibrating Rod Level Switch
Application	S	Solids
Mounting Type	IN/RE	Integral / Remote
Probe Type	RP/HF	Rigid Pipe / Hose Flexible
Insulation Coating	NI/PT	Non-Insulated /PTFE Insulated
Probe Diameter	D1	Standard 44 mm
Probe Material	04 /4L/ 16/ 6L/MS/ SM	SS304/SS304L / SS316/ SS316L /Mild steel/ Special material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED
Enclosure	WP / FL	Weatherproof / Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B1 / B2 / B3/B4	1" / 1½" / 2" BSP/ Special Mounting
	N1 / N2 / N3/N4	1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face
MOC-Process Connection	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customized
Temperature	T1	Max -20–150°C
Remote Cable	05 / SR	5 /Special Customer Requirement
Cable Gland	G1/ G2 / G3	PG13.5 / ½"NPT / ¾" ET
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	XXXX	260 mm TO Customized

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Mechanical Drawing:



INFORMATION REQUIRED Before Purchase

- Process Parameters: Process Connection, Process Temperature, Vessel pressure if any.
- Application media characteristics: Density, Viscosity, Sp. Gr.
- Supply Voltage, Probe length, Mounting etc.

Manufactured By:

Levtron Instruments Pvt Ltd.

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